

Weiting Liu

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Personal Website <https://vickyliuwt.github.io/> | U.S. Citizen (No sponsorship required)

EDUCATION

Northeastern University — Boston, MA | Sep. 2024 – Expect May 2027

Master of Science in Computer Science, GPA: 4.0/4.0

Courses: Software Development, Algorithms, Database Systems, Web Technologies, Machine Learning

Savannah College of Art and Design — Atlanta, GA | Jun. 2023

Bachelor of Fine Arts in Animation

Courses: Character Animation, Tech Animation, Programming, Life Drawing, Collaborative Projects

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, SQL

Frontend: React, Angular, Vue.js, Redux, HTML5, CSS3, Tailwind CSS

Frameworks & APIs: Spring Boot, Node.js, Express.js, RESTful APIs, FastAPI

Databases & Messaging: PostgreSQL, MySQL, Redis, MongoDB, Kafka

Cloud & DevOps: AWS (EC2, S3, Lambda, EKS), Docker, Kubernetes, CI/CD, GitHub Actions, Git

Design & Creative Tools: Adobe Creative Cloud, Autodesk Maya, Blender, Figma

Tools & Process: Agile/Scrum, Postman, JUnit, pytest

WORK EXPERIENCE

Huibo Robotics Technology Co., Ltd. — Jiangsu, China | May 2025 – Sept. 2025

Software Engineer Intern | **React, Spring Boot, PostgreSQL, Docker, CI/CD, WebSocket**

- Designed a real-time robot status monitoring dashboard with **React** and **WebSocket**, visualizing joint angles, torque, and error states across fleets.
- Implemented **RESTful APIs** with **Spring Boot** to expose telemetry and control data, enabling downstream clients to query and visualize robot logs.
- Stored robot telemetry logs and event records in **PostgreSQL** with time-series indexing, reducing historical query latency by 45% across dashboards.
- Automated deployment of monitoring services with **Docker** and **GitHub Actions CI/CD** pipelines, ensuring reproducible builds across environments.

Deluxeton Homes LLC — GA | Jun. 2023 – Aug. 2024

Interior Designer

- Developed **Python** automation scripts for material cost analysis and reporting, reducing documentation effort across interior design projects.
- Created detailed CAD drawings, floor plans, and 3D models using Autodesk tools, translating client requirements into visual design specifications.
- Collaborated with clients and stakeholders to present design concepts, gathering feedback and refining proposals through iterative communication.

Deluxeton Homes LLC — GA | May 2022 – Aug. 2022

Web Developer | **HTML, CSS, JavaScript, Node.js, MongoDB, Bootstrap**

- Developed responsive web pages with **HTML, CSS**, and **Bootstrap**, implementing color systems and typography from design specs across interfaces.
- Developed reusable page templates and UI components with **JavaScript**, standardizing shared elements to improve consistency across multiple pages.
- Built back-end functionality with **Node.js** and **MongoDB**, implementing APIs and data models to support dynamic web content and product data.
- Coordinated with the product and business team on web asset art direction, ensuring visual brand consistency across all layouts and digital deliverables.

Dezure Development — GA | May 2021 – Aug. 2021

Architectural Intern

- Produced concept sketches and designs from client requirements, translating briefs into visual proposals for architectural review and approval.
- Researched construction materials, dimensions, and specifications, identifying potential structural issues during early design and planning phases.

First Fine Art & Design Academy — GA | May 2020 – Aug. 2020

Teaching Assistant

- Provided 1-on-1 guidance to students on technical animation concepts, identifying individual learning gaps and adapting instructional approaches.
- Assisted professors in developing materials and managing classroom engagement across group and individual art technique learning sessions.

PROJECTS

Secure Reliable File Transfer | **Python, Raw Sockets (IP/UDP), AES-256-GCM, HMAC/HKDF-SHA256, Multithreading, AWS EC2**

- Engineered a reliable transport protocol from scratch over raw UDP sockets, hand-crafting IP, UDP, and a custom header, delivering files with byte-for-byte SHA-256/MD5 integrity across AWS EC2.
- Implemented Go-Back-N reliability on a multithreaded send/ACK/retransmit architecture, sustaining zero-corruption delivery under packet loss.
- Secured every datagram with a PSK handshake, HKDF-SHA256, and AES-256-GCM authenticated encryption binding packet metadata as AAD.
- Hardened the protocol with replay protection and end-to-end hash verification, passing 5/5 adversarial tests.

Quora Duplicate Question Detection | **81.67% Accuracy across 3 ML Approaches** | **Python, scikit-learn, PyTorch, NLTK, Gensim**

- Built and benchmarked three duplicate-detection models on 404K+ question pairs, achieving 81.67% accuracy and 0.90 AUC-ROC over majority-class.
- Engineered an NLP preprocessing pipeline and 5 handcrafted similarity features fused with 50K features for the top Logistic Regression classifier.
- Trained a Siamese LSTM in PyTorch with frozen GloVe embeddings and shared weights, hitting 79.2% accuracy and the highest recall.
- Compared Word2Vec vs GloVe sentence embeddings and shipped a standalone demo.py and an interactive CLI supporting all three models.

Kambaz LMS – Full-Stack Learning Management System | **Next.js, React, TypeScript, Redux Toolkit, Node.js, Express, MongoDB Atlas**

- Developed a full-stack Canvas-style LMS with role-based access governing faculty and student capabilities.
- Built RESTful CRUD APIs with Node.js/Express backed by MongoDB Atlas, modeling users, courses, and enrollments for persistent multi-user state.
- Implemented client-side state management with Redux Toolkit and role-guarded routing, conditionally rendering instructor and student experiences.
- Styled a responsive React-Bootstrap UI and deployed frontend and backend to production with Git-based version control.

Memer – Native iOS Meme Generator & Social App | **Swift, UIKit, Firebase Auth, Cloud Firestore, Firebase Storage, MVC**

- Built a native iOS meme generator and social app in Swift/UIKit on an MVC architecture, spanning 15+ screens
- Engineered a canvas-style editor with draggable, pinch-scalable, flattening layers into a single exported image via UIGraphicsImageRenderer.
- Integrated Firebase Auth, Cloud, and Storage for authentication, real-time reactions, and friend-to-friend chat powered by Firestore snapshot listeners.
- Implemented camera and photo-library capture and a SessionManager caching layer for fast profile loads.